

23AID472 Text Analytics

Duplicate Assignment & Essay Detection via Semantic Similarity

Zeroth Review

Group 9 (AID - B)

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1. Introduction & Motivation

- Academic integrity relies heavily on the authentic submission of assignments and essays.
- Digital assignment submissions have scaled drastically, making manual cross-comparison for duplicate submissions virtually impossible for instructors.
- There is an urgent institutional need for automated, intelligent pipelines that can scan batches of student texts and index their underlying conceptual content.

2. Problem Definition

The Core Vulnerability

Traditional text matching algorithms (e.g., character-level or string-matching patterns) rely heavily on exact lexical overlap. They fail entirely when a student restructures an essay using synonym substitutions or syntax swapping to preserve identical meanings.

The Technical Gap:

- String matchers classify "The automobile is rapid" and "The vehicle moves fast" as entirely unique.
- Our core problem is tracking **semantic duplicity** rather than simple vocabulary copying.

3. Project Objectives

This project focuses on building an automated Text Analytics pipeline that achieves the following goals:

- **Semantic Mapping:** Projecting arbitrary lengths of student text into a continuous vector space where thematic intent dictates alignment.
- **Paraphrase Resilience:** Designing a matching layer that successfully identifies plagiarized ideas even when masked by heavy rephrasing.
- **Comparative Framework:** Demonstrating a progression from classic bag-of-words counting methods to deep neural representations.

4. Dataset Strategy

To train, validate, and prove the efficacy of our semantic matching approach, we will utilize established open-source paraphrase corpora that reflect student rephrasing styles:

A. Microsoft Research Paraphrase Corpus (MSRP)

- Consists of thousands of sentence pairs systematically extracted from multi-source web news engines.
- Each pair is manually annotated by human experts to ensure true semantic equivalence and structural intent.
- Excellent for evaluating how robustly our system handles highly complex, structurally altered sentences.

B. Quora Question Pairs (QQP)

- A massive dataset containing over 400,000 ground-truth labeled sentence pairs.
- Labeled explicitly to distinguish binary semantic duplication (whether two questions have the same core intent).
- Forces models to learn alternative grammatical and lexical paths that lead to the exact same core idea.

5. Proposed Idea: The Siamese Modeling Approach

Instead of calculating metrics sequentially, our core proposed architecture introduces a **Siamese Neural Network layout**.

- **Twin Encoding Columns:** Document A and Document B are passed concurrently into two independent extraction channels.
- **Shared Weight Paradigm:** Both channels share identical model parameters, forcing both inputs to be parsed under the exact same semantic constraints.
- **Geometric Vector Output:** The text inputs are converted into uniform, fixed-size dense coordinate representations for geometric comparison.

6. Idea of Approach: Similarity Evaluation

Once the Siamese encoding blocks map the text pairs into their respective dense vectors ($\mathbf{u}$ and $\mathbf{v}$), the final engine calculates a normalized similarity metric:

Mathematical Distance Matching

The system will use **Cosine Similarity** to calculate the angular proximity between document profiles:

$$\text{Similarity}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

- A threshold value τ (e.g., 0.8) will divide text inputs into binary classes: **Unique** vs. **Duplicate / Flagged**.

7. Architectural Comparison

Our proposed approach traces the progression of Text Analytics paradigms to validate system choices:

- **Phase 1 Baseline (Sparse Representation):** Implement a statistical **TF-IDF + Cosine Space** setup. This establishes a baseline for detecting raw keyword copying.
- **Phase 2 Intermediate (Static Dense Space):** Transition to a **Word2Vec Document Vector Average** layer. This begins catching simple single-word synonym replacements.
- **Phase 3 Target (Contextual Deep Space):** Deploy a full **Transformer Sequence Encoder** to capture the complete ordering, syntax, and long-range semantic dependency within the essays.

8.Project Workflow

The system operational flow is divided into four distinct systemic modules:

System Pipeline Architecture

- ➊ **Ingestion & Preprocessing:** Lowercasing, noise filtering, stop-word reduction, and word root lemmatization.
- ➋ **Vector Representation:** Passing normalized text streams into the twin vector layers.
- ➌ **Metric Extraction:** Applying distance formulas to compile semantic matching maps.
- ➍ **Decision Engine:** Classifying pairs based on threshold margins and highlighting duplicate sections.

9. Expected Deliverables & Roadmap

- **Weeks 1–4 (Pipeline Framework):** Consolidating preprocessing pipelines and establishing our sparse baseline models.
- **Weeks 5–8 (Dense Space Implementation):** Incorporating static embeddings and pooling vectors to catch structural vocabulary variations.
- **Weeks 9–12 (Deep Model Integration):** Deploying sequence transformers within the Siamese matching core.

Thank You!